

Project Report – Automated Network Request Management System

1. INTRODUCTION

1.1 Project Overview

Organizations often manage network service requests such as VPN access, firewall rule changes, VLAN creation, IP address allocation, switch port activation, DNS configuration, and network device setup through manual processes involving emails, phone calls, spreadsheets, or traditional ticketing methods. These approaches frequently lead to delayed approvals, inconsistent request handling, poor communication between users and the Network Operations team, and limited visibility into the request lifecycle. Employees are often required to follow up manually to determine the status of their requests, while network administrators spend considerable time coordinating approvals, assigning tasks, and updating request progress.

To address these challenges, this project presents the **Automated Network Request Management System Using ServiceNow**, a web-based application developed on the ServiceNow platform. The application automates the complete lifecycle of network service requests by providing a centralized workflow for employees, managers, network engineers, and system administrators. Employees can submit network service requests through the Service Catalog, while managers review and approve requests based on organizational policies. Upon approval, the system automatically creates implementation tasks, assigns them to the appropriate Network Engineer, and tracks the implementation process until successful completion.

The application implements role-based access control using ServiceNow roles, ensuring that users can access only the information and functionalities assigned to their respective roles. Workflow automation is achieved through ServiceNow Flow Designer, which automatically manages approvals, task creation, assignment, status updates, and email notifications throughout the request lifecycle. This significantly reduces manual effort, improves communication, enhances transparency, and ensures efficient management of network service requests.

The proposed solution provides a scalable, secure, and centralized platform that improves operational efficiency, enhances user experience, and standardizes the network request management process for enterprise IT environments.

1.2 Purpose

The primary purpose of the **Automated Network Request Management System Using ServiceNow** is to replace traditional manual network request processing with a centralized and automated solution developed on the ServiceNow platform. The system enables employees to request network services digitally while providing managers and network engineers with a structured workflow for approval, implementation, and request tracking.

The specific objectives of the project are:

- To provide a centralized platform for submitting and managing network service requests.
- To automate the network request workflow using ServiceNow Flow Designer.
- To enable managers to review, approve, or reject submitted network requests.
- To automatically generate implementation tasks and assign them to the Network Operations team.
- To provide role-based access control for employees, managers, network engineers, and system administrators.
- To send automated email notifications whenever the request status changes.
- To reduce manual intervention, improve transparency, and enhance communication throughout the request lifecycle.
- To demonstrate the practical implementation of ServiceNow development concepts, including scoped applications, Service Catalog, custom tables, Flow Designer, Business Rules, Catalog UI Policies, Client Scripts, Assignment Rules, Notifications, Reports, and Dashboards.

2. PROJECT OVERVIEW

2.1 Problem Statement

2.1 PROBLEM STATEMENT

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1  Employee (Requester)	Request network services such as VPN access, firewall changes, VLAN creation, or IP allocation quickly and efficiently.	The request process is manual and requires multiple emails or phone calls before reaching the appropriate team.	There is no centralized system to submit, track, or monitor network requests.		 Frustrated, uncertain, and delayed in completing my work.
PS-2  Manager (Approver)	Review and approve network service requests efficiently.	Requests arrive from multiple communication channels without proper tracking or prioritization.	Manual processing makes it difficult to manage approvals consistently and on time.		 Overloaded, inefficient, and concerned about delayed service delivery.
PS-3  Network Engineer	Receive implementation tasks with complete and accurate request information.	Task assignments are performed manually and may contain incomplete or inconsistent details.	There is no automated workflow to generate and assign implementation tasks.		 Confused, delayed, and less productive while fulfilling requests.
PS-4  IT Administrator	Monitor the overall network request process and maintain audit records.	Request information is scattered across emails and spreadsheets, making reporting and auditing difficult.	Manual tracking does not provide centralized visibility or historical records.		 Concerned about operational efficiency, compliance, and service quality.

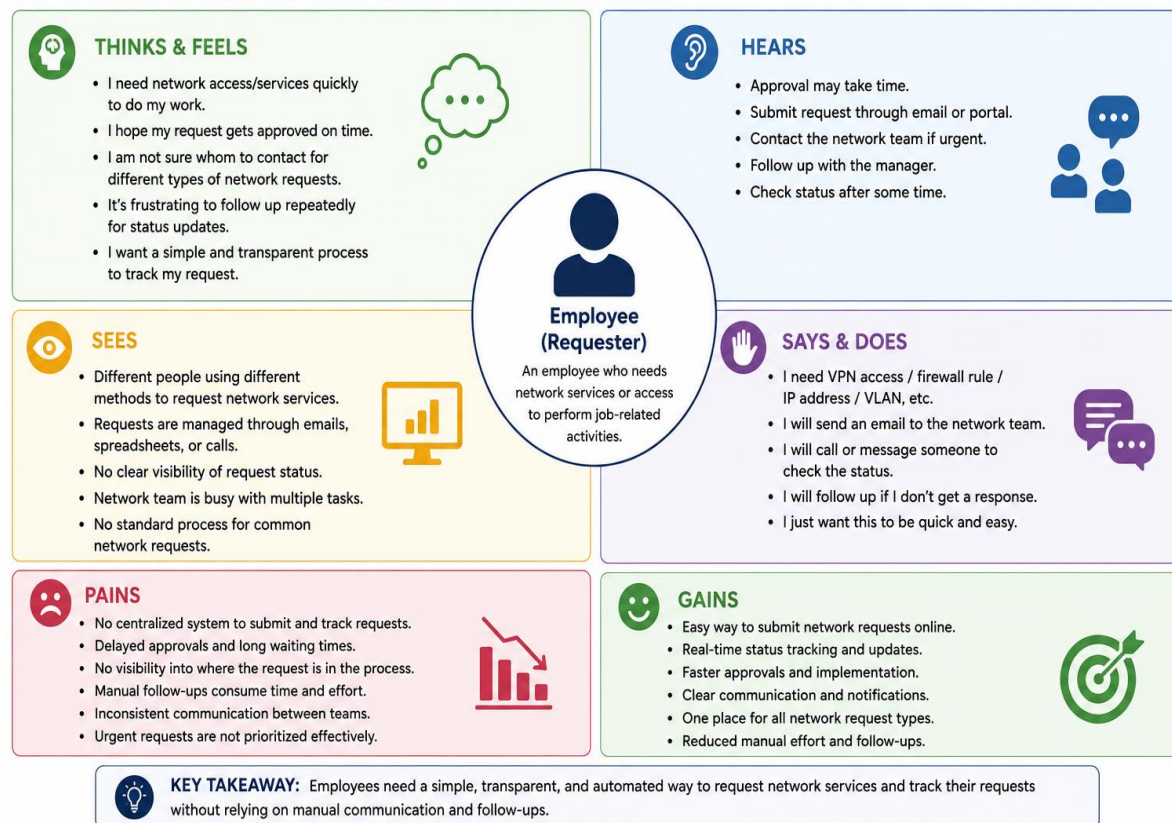
Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Employee (Requester)	Request network services such as VPN access, firewall changes, VLAN creation, or	The request process is manual and requires multiple emails	There is no centralized system to submit, track, or monitor	Frustrated, uncertain, and delayed in

		IP allocation quickly and efficiently	or phone calls before reaching the appropriate team.	network requests.	completing my work.
PS-2	Manager (Approver)	Review and approve network service requests efficiently..	Requests arrive from multiple communication channels without proper tracking or prioritization.	Manual processing makes it difficult to manage approvals consistently and on time.	Overloaded, inefficient, and concerned about delayed service delivery.

2.2 Empathy Map Canvas

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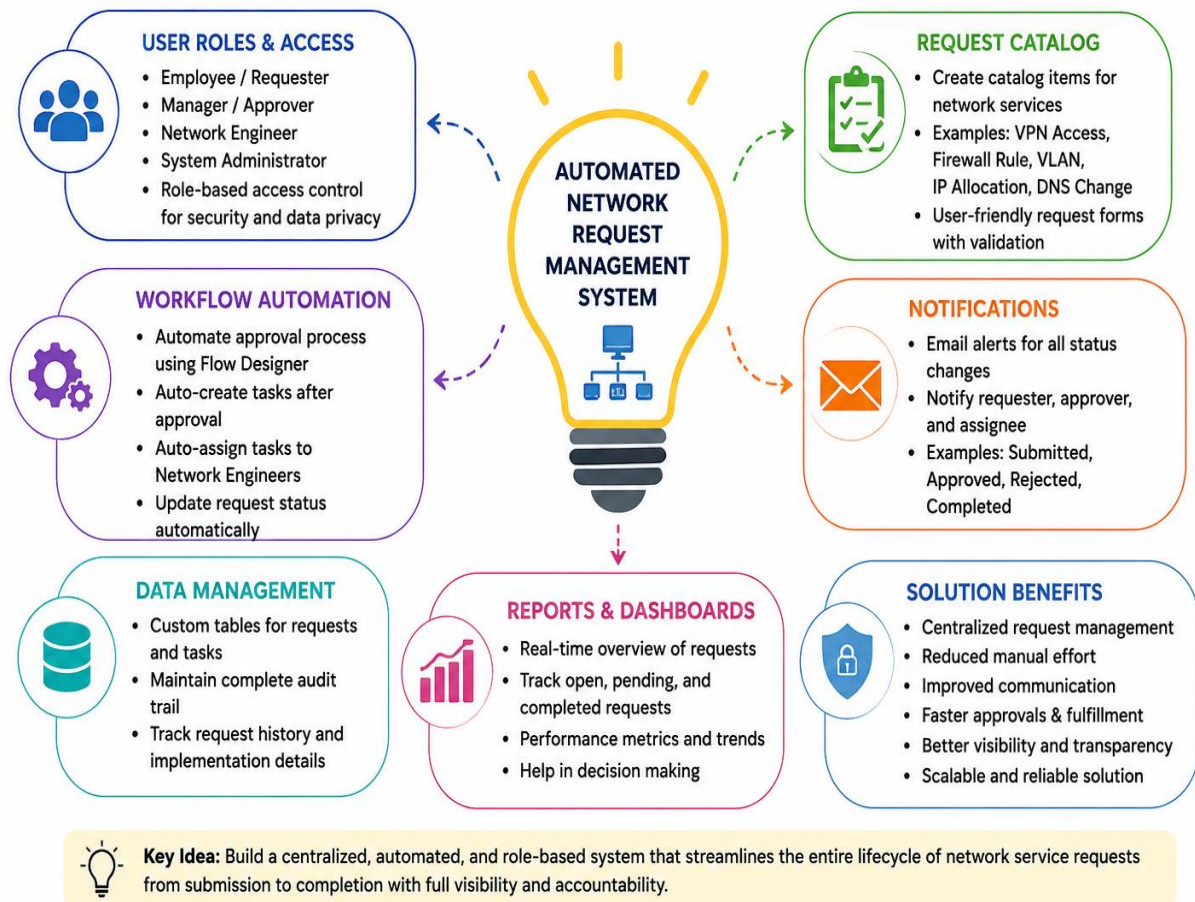
Project: Automated Network Request Management System Using ServiceNow



2.3 Brainstorming

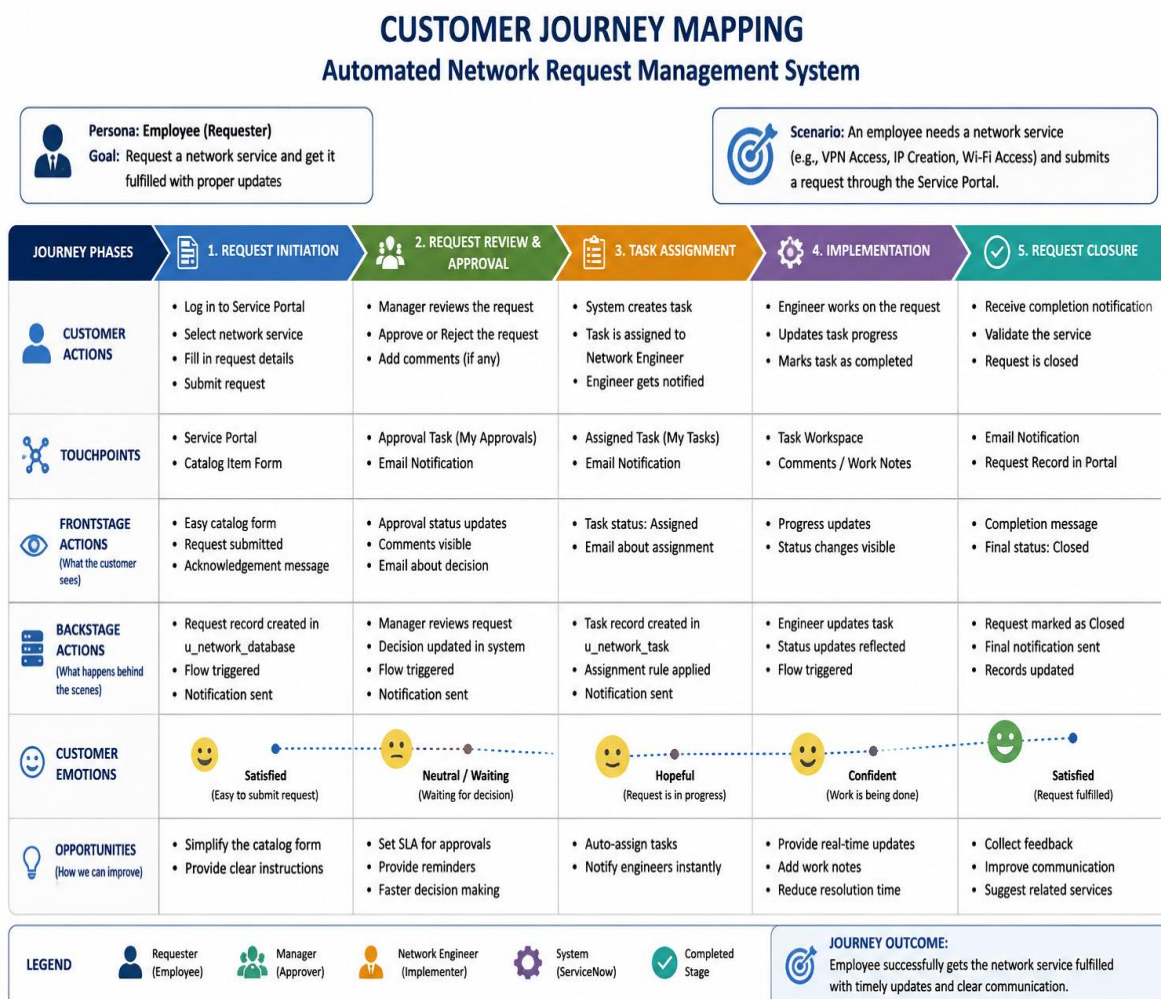
2.3 BRAINSTORMING

The following ideas and inputs were considered during the brainstorming phase to design the **Automated Network Request Management System**.



3. REQUIREMENT ANALYSYS

3.1 Customer Journey Map



3.2 Functional Requirements

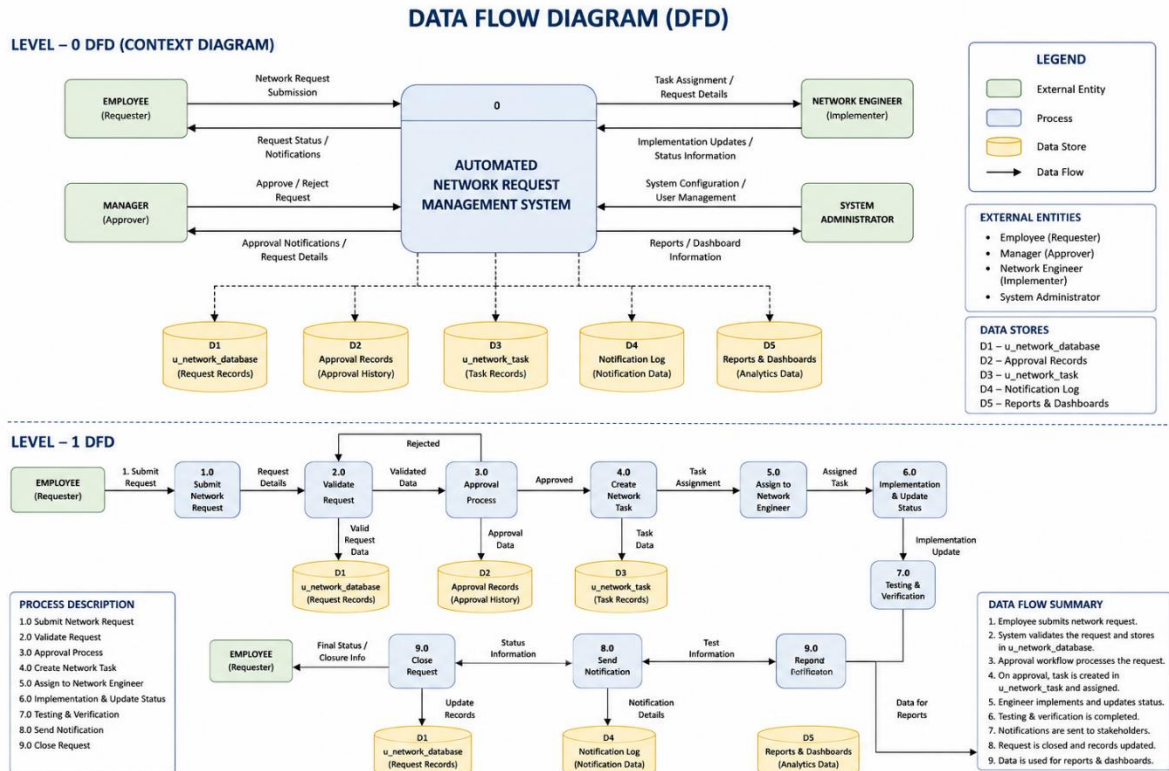
Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication & Role Management	Authenticate users using the existing ServiceNow Users (sys_user) table. Assign Employee role to requesters. Assign Manager role for approvals. Assign Network Engineer role for implementation tasks. Assign System Administrator role for application management.
FR-2	Network Request Management	Employees can submit network service requests through the Service Catalog. Capture request details using catalog variables. Create request records in u_network_database. Validate mandatory request information before submission.
FR-3	Approval Management	Managers can review submitted network requests. Approve network requests. Reject network requests with comments. Automatically notify users of approval decisions.
FR-4	Task Assignment & Implementation	Automatically create fulfillment tasks in u_network_task after approval. Assign tasks to the Network Operations team using assignment rules. Allow Network Engineers to update implementation progress and completion status.
FR-5	Workflow Automation & Notifications	Trigger Flow Designer workflows after request submission and approval. Send email notifications for request creation, approval, rejection, task assignment, and completion. Automatically update request status throughout the lifecycle. Generate reports and dashboards for monitoring network requests.

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application shall provide a simple, user-friendly Service Portal interface for employees, managers, and network engineers to efficiently submit, review, and manage network service requests.
NFR-2	Security	Access shall be controlled using ServiceNow role-based access control (RBAC). Users can only access records and modules permitted by their assigned roles.
NFR-3	Reliability	The system shall accurately store all network request records, approval details, and task information while ensuring workflow execution without data loss.
NFR-4	Performance	Network requests, approvals, task creation, notifications, and status updates shall be processed with minimal response time under normal operating conditions.
NFR-5	Availability	The application shall remain available whenever the ServiceNow instance is operational, ensuring uninterrupted access to network request services.
NFR-6	Scalability	The system shall support increasing numbers of users, network requests, and fulfillment tasks without requiring significant architectural modifications.

3.3 Data Flow Diagram: A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.



3.4 Technology Stack

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Employees, Managers, and Network Engineers interact with the application through the Service Portal and request forms.	ServiceNow Service Portal, Catalog Item, Forms, Lists
2	Application Logic	Processes network request validation, request lifecycle management, and business logic execution.	Business Rules, Script Includes

S.No	Component	Description	Technology
3	Workflow Automation	Automates request approvals, task creation, status updates, and notifications.	Flow Designer
4	Database	Stores user information, network request records, implementation tasks, and request history.	sys_user, u_network_database, u_network_task
5	Notification Service	Sends automated notifications for request submission, approval, rejection, task assignment, and request completion.	ServiceNow Notifications, Email Templates
6	Security & Access Control	Provides secure access using role-based permissions for employees, managers, engineers, and administrators.	ACLs, Roles, User Criteria
7	Cloud Infrastructure	Hosts and manages the application on the ServiceNow cloud platform with centralized data management.	ServiceNow Cloud Platform

Table 2: Application Characteristics

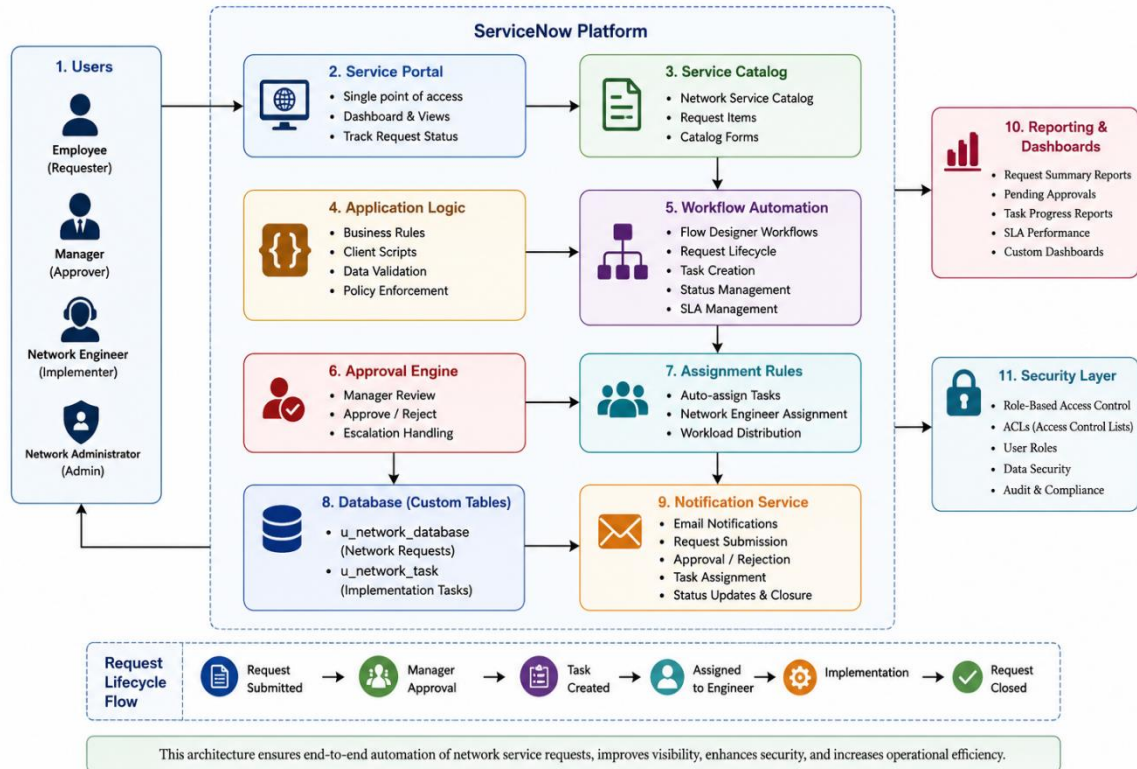
Characteristic	Description	Technology
Security	Provides secure authentication and role-based authorization for employees, managers, network engineers, and administrators.	ACLs, Roles, User Criteria
Scalability	Supports increasing numbers of network requests, users, and implementation tasks without significant performance degradation.	ServiceNow Platform
Availability	Ensures continuous access to the application whenever the ServiceNow instance is operational.	ServiceNow Cloud
Performance	Delivers efficient request processing through automated workflows, optimized forms, and background execution.	Flow Designer, Business Rules
Reliability	Maintains accurate request records, task information, approvals, and notifications while ensuring consistent workflow execution.	ServiceNow Database, Flow Designer

4. PROJECT DESIGN

4.3 Solution Architecture

Architecture Components

The architecture of the Automated Network Request Management System Using ServiceNow consists of the following components:



4.1 Problem–Solution Fit

Section	Content
1. Customer Segment(s)	• Employees (Requesters)• Managers (Approvers)• Network Engineers• System Administrators
2. Jobs-to-be-done / Problems	• Submit network service requests quickly. • Obtain timely approval for network requests. • Track request status in real time. • Reduce manual communication and paperwork. • Enable efficient task assignment and implementation.
3. Triggers	• Employee requires VPN access for remote work. • Firewall rule modification is needed. • New VLAN or IP address allocation is requested. • Network device configuration or maintenance is required. • Network Operations team receives a new network request.
4. Emotions (Before / After)	Before: Frustrated, uncertain, delayed, and dependent on manual follow-ups. After: Confident, informed, satisfied, and able to track requests with transparency.

Section	Content
5. Available Solutions	• Email-based request submission. • Phone calls to the Network Operations team. • Spreadsheet-based request tracking. • Manual approval process. • Traditional IT Help Desk support.
6. Customer Constraints	• No centralized request management system. • Delayed approvals and task assignment. • Lack of request visibility and tracking. • Inconsistent communication between users and engineers. • Difficulty maintaining audit records.
7. Behaviour	• Employees submit requests through emails or phone calls. • Managers manually review and approve requests. • Network Engineers receive tasks through different communication channels. • Employees repeatedly follow up for status updates.
8. Channels of Behaviour	Online: ServiceNow Service Portal, Service Catalog, Email Notifications, Dashboards. Offline: Phone calls, Internal meetings, Face-to-face communication with the IT team.
9. Problem Root Cause	The organization lacks a centralized and automated platform for managing network service requests. Requests are processed manually through multiple communication channels, resulting in delayed approvals, inconsistent task assignments, poor visibility, and inefficient request management.
10. Your Solution	Develop an Automated Network Request Management System Using ServiceNow that enables employees to submit network service requests through the Service Catalog. The system automatically validates requests, routes them for approval, creates implementation tasks, assigns Network Engineers, sends email notifications, tracks request status, and maintains complete audit records using custom ServiceNow workflows and tables.

4.2 Proposed Solution

S.No	Parameter	Description
1	Problem Statement (Problem to be solved)	Organizations often process network service requests through emails, phone calls, or manual ticketing systems. These traditional methods lead to delayed approvals, poor communication, lack of request visibility, inconsistent task assignment, and difficulty in tracking request progress and maintaining audit records.
2	Idea / Solution Description	Develop an Automated Network Request Management System Using ServiceNow where employees submit network service requests through the Service Catalog. The system validates request information, routes requests to managers for approval, automatically creates implementation tasks for approved requests, assigns Network Engineers using assignment rules, sends automated email notifications, and enables users to track request progress until completion.
3	Novelty / Uniqueness	The solution combines Service Catalog, Flow Designer, automated approvals, task automation, assignment rules, role-based access control, email notifications, real-time tracking, reporting, and custom ServiceNow

S.No	Parameter	Description
		tables within a single integrated application. It eliminates manual request handling while providing transparency, auditability, and standardized network service management.
4	Social Impact / Customer Satisfaction	The application improves operational efficiency by reducing manual effort, accelerating request processing, improving communication among employees and the Network Operations team, increasing transparency, and providing real-time updates throughout the network request lifecycle.
5	Business Model (Revenue Model)	The solution is designed as an enterprise IT Service Management application for organizations using the ServiceNow platform. It can be deployed as an internal network request automation solution and extended as a reusable ServiceNow application or subscription-based enterprise service for multiple organizations.
6	Scalability of the Solution	Built on the ServiceNow cloud platform, the application can support increasing numbers of users, departments, network requests, and implementation tasks. Future enhancements may include CMDB integration, SLA management, AI-assisted request routing, predictive analytics, network monitoring integration, and advanced reporting dashboards without major architectural changes.

5. PROJECT PLANNING & SCHEDULING

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority
Sprint-1	Project Setup	USN-1	Create scoped application, roles, Leave Request table, Student View, leave submission	8	High
Sprint-2	Approval Workflow	USN-2	Faculty View, Class Teacher approval, rejection, manual escalation to HOD	9	High
Sprint-3	HOD and Notifications	USN-4	HOD approval, modify leave days, Flow Designer, email notifications	8	Medium
Sprint-4	Testing and Documentation	USN-5	Testing, bug fixes, reports, final documentation, presentation	5	High

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	8	2 Days	22 June 2026	23 June 2026	8	23 June 2026
Sprint-2	9	2 Days	24 June 2026	25 June 2026	9	25 June 2026
Sprint-3	8	2 Days	26 June 2026	27 June 2026	8	27 June 2026
Sprint-4	5	1 Day	28 June 2026	28 June 2026	5	28 June 2026

Velocity:

- **Total Story Points: 30**
- **Total Duration: 7 days**

Average Velocity = $30 \div 7 \approx 4.3$ story points/day

6. FUNCTIONAL AND PERFORMANCE TESTING

2. Milestone 1 – Service Catalog Setup

Objective

Create the Service Catalog and initialize the network request management system.

Activities Performed

- Created the Network Services catalog and category.
- Developed catalog items for network service requests.
- Configured catalog variables and prepared the development environment.

Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated
3M Privacy Filter - Lenovo X1 Carbon	Privacy Filter - X1 Carbon	true		Service Catalog	Peripherals	\$43.19	Item	2022-11-20 20:46:33
3M Privacy Filter - MacBook Pro	Privacy Filter	true		Service Catalog	Peripherals	\$42.23	Item	2022-11-20 20:46:33
3M Privacy Filter - MacBook Pro Retina	Privacy Filter	true		Service Catalog	Peripherals	\$40.31	Item	2022-11-20 16:00:00
☐ Access	Microsoft Access	true		Service Catalog	Software	\$139.99	Item	2022-11-20 20:46:33
Acrobat	Adobe Acrobat	true		Service Catalog	Software	\$139.99	Item	2022-12-05 20:46:33
Add network switch to datacenter cabinet	This standard change template describes ...	true		Service Catalog	Network Standard Changes	\$0.00	Item	2026-05-04 18:21:03
Add/Remove users from group	Add/Remove users from group	true		Service Catalog	Services	\$0.00	Item	2022-12-05 10:17:33

3. Milestone 2 – Database Configuration

Objective

Create the custom database tables for storing network request details and task information.

Activities Performed

- Designed the u_network_database table to store network service requests.
- Created the u_network_task table to manage request fulfillment tasks.
- Added fields such as Request Number, Requested By, Request Type, Department, Priority, Status, Assigned To, Due Date, and Comments.
- Configured reference fields linked to the sys_user table for users and assignees.

Table Network Request

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label

* Name

Application

Remote Table

Columns Controls Application Access

Table Columns for text Search

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Created	Date/Time	(empty)		40	false
Comments	String	(empty)		500	false
Requested For	Reference	User		32	false
Updated by	String	(empty)		40	false
Sys ID	Sys ID (GUID)	(empty)		32	false
Business Justification	String	(empty)		255	false
Updates	Integer	(empty)		40	false
Priority	Choice	(empty)		40	false
Updated	Date/Time	(empty)		40	false
Requested Date	Date/Time	(empty)		40	false

Milestone 3 – User & Role Configuration

Objective

Configure application users, user groups, and role-based access control for the Automated Network Request Management System.

Activities Performed

- Created Employee (Requester) user accounts.
- Created Manager (Approver) user accounts.
- Created Network Engineer user accounts.
- Created System Administrator user accounts.
- Configured Network Operations assignment group. Assigned Employee role to users for submitting network service requests.
- Assigned Manager role for reviewing and approving requests.
- Assigned Network Engineer role for implementing approved network tasks.
- Assigned System Administrator role for managing application configuration, workflows, and user access.
- Verified role-based permissions and access to Service Portal, Service Catalog, custom tables, and workflow components.

meinda.carleton	Meinda Carleton	meinda.carleton@example.com	true	2012-02-17 19:04:49	2026-05-04 18:29:51
jewel.agresta	Jewel Agresta	jewel.agresta@example.com	true	2012-02-17 19:04:49	2026-05-04 18:29:51
sean.bonnet	Sean Bonnet	sean.bonnet@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
jacinto.gawron	Jacinto Gawron	jacinto.gawron@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
krystle.stika	Krystle Stika	krystle.stika@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
billie.cowley	Billie Cowley	billie.cowley@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
christian.marnell	Christian Marnell	christian.marnell@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
naomi.greenly	Naomi Greenly	naomi.greenly@example.com	true	2012-02-17 19:04:53	2026-05-04 18:29:51
jess.assad	Jess Assad	jess.assad@example.com	true	2012-02-17 19:04:49	2026-05-04 18:29:51
cherie.fuhri	Cherie Fuhri	cherie.fuhri@example.com	true	2012-02-17 19:04:49	2026-05-04 18:29:51
geri.forness	Geri Forness	geri.forness@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
nelly.jakuboski	Nelly Jakuboski	nelly.jakuboski@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
dwain.cuttitta	Dwain Cuttitta	dwain.cuttitta@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
justina.dragaj	Justina Dragaj	justina.dragaj@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
allyson.gillispie	Allyson Gillispie	allyson.gillispie@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
bridget.knightly	Bridget Knightly	bridget.knightly@example.com	true	2012-02-17 19:04:50	2026-05-04 18:29:51
certification_user	Certification User	certification@example.com	true	2012-10-02 11:40:54	2026-05-04 18:29:51
dollie.pillitteri	Dollie Pillitteri	dollie.pillitteri@example.com	true	2012-02-17 19:04:53	2026-05-04 18:29:51
elmo.gabouer	Elmo Gabouer	elmo.gabouer@example.com	true	2012-02-17 19:04:53	2026-05-04 18:29:51

5. Milestone 4 – User Interface Development

Objective

Develop intuitive and user-friendly interfaces for employees, managers, and network engineers to efficiently submit, manage, and track network service requests.

Activities Performed

- Designed the Network Service Request Catalog Item for submitting network requests.
- Configured catalog variables to capture request details such as Request Type, Priority, Business Justification, and Configuration Information.
- Developed user-friendly request forms using the ServiceNow Service Catalog.
- Implemented Catalog UI Policies to provide dynamic form behavior by showing, hiding, and making fields mandatory based on the selected request type.
- Configured Catalog Client Scripts to perform client-side validation and improve user interaction.
- Designed separate interfaces for request submission and request fulfillment based on user roles.
- Customized form layouts to improve usability and simplify data entry.
- Validated the user interface to ensure a consistent and responsive user experience across different request scenarios.

Build and modify items faster with the improved [Catalog Builder](#).

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name: VPN Access Request

Application: Global

Catalogs: Network Services

Category: Access Requests

State: -- None --

Checked out: -- None --

Owner: System Administrator

Active: ☒

Fulfillment automation level: Unspecified

6. Milestone 5 – Workflow Automation

Objective

Automate the complete lifecycle of network service requests, including request processing, approval, task creation, assignment, notifications, and request closure using ServiceNow Flow Designer.

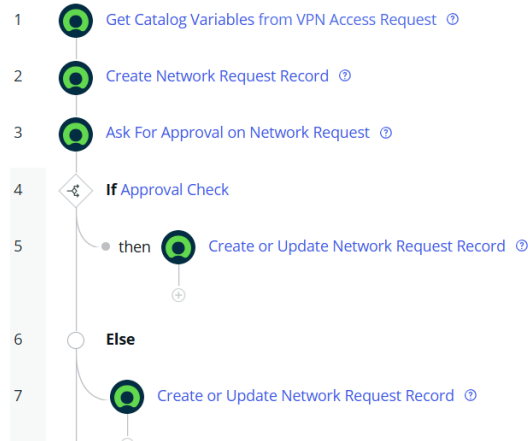
Activities Performed

- Developed an automated workflow using ServiceNow Flow Designer.
- Configured the workflow to trigger automatically when a network service request is submitted.
- Implemented an approval process for requests requiring managerial authorization.
- Automated the creation of implementation tasks for approved network requests.
- Configured assignment rules to automatically assign tasks to the Network Operations team.
- Implemented automated email notifications for request submission, approval, rejection, task assignment, and request completion.
- Configured automatic request status updates throughout the workflow lifecycle.
- Validated the end-to-end workflow to ensure successful automation from request submission to closure.

TRIGGER



ACTIONS



ACTIONS

1	Get Catalog Variables from VPN Access Request	Core Action	Completed	2026-06-30 05:05:51	246ms
2	Create Record	Core Action	Completed	2026-06-30 05:05:51	18ms
3	Ask For Approval	Core Action	Waiting	2026-06-30 05:05:51	770ms
4	If Approval Check	Flow Logic	Not Run		
5	Create or Update Record	Core Action	Not Run		
6	Else	Flow Logic	Not Run		
7	Create or Update Record	Core Action	Not Run		

Test Scenario 1 – Request Submission

Objective: Verify that an email is generated when a network request is submitted.

Steps:

1. Login as an Employee.
2. Open the Service Portal.
3. Submit a **VPN Access** request.
4. Click **Submit**.

Expected Result:

- Request is created successfully.
- Status changes to **Pending Approval**.
- A submission email notification is generated.

Preview Email

```
Job Name: Daily job to fetch Email Indicator Data and Email Notifications created
Run time: 1 Second
Inserts: 0
Deletes: 0
Warnings: 0
Errors: 1

Unsubscribe [mailto:?subject=Unsubscribe from "PA collection job with error"&body=Sending this email with the
predefined content in the subject and body will unsubscribe you from the notification "PA collection job with
error"%0D%0A%0D%0AUnsubscribe:{"id":"db8e5120d7330100fa6c0c12ce610309","token":"c797e62132"}] | Notification
Preferences [https://dev306160.service-now.com/nav_to.do?uri=notification_preferences.do%3Fsysparm_notification=db8e5120d7330100fa6c0c12ce610309]

Ref:MSG0001997_aRg2tfsURWp4MPUK0htY
```

8. ADVANTAGES AND DISADVANTAGES

Advantages

1. Centralized Network Request Management

Provides a single platform for employees to submit, track, and manage all network service requests efficiently.

2. Automated Workflow

Automates the complete request lifecycle, including approvals, task creation, assignment, notifications, and request closure using ServiceNow Flow Designer.

3. Role-Based Access Control

Implements secure role-based access, allowing employees, managers, network engineers, and administrators to access only the functionalities relevant to their roles.

4. Real-Time Request Tracking

Employees can monitor the status of their network service requests throughout the approval and implementation process.

5. Automated Email Notifications

The system automatically sends email notifications for request submission, approval, rejection, task assignment, progress updates, and request completion.

6. Improved Operational Efficiency

Reduces manual processing and accelerates network service delivery through workflow automation and automatic task assignment.

7. Centralized Task Management

Automatically generates implementation tasks and assigns them to the appropriate Network Engineer, improving coordination and accountability.

8. Secure Data Management

Network request information is securely stored within the ServiceNow platform using custom tables and protected through Access Control Lists (ACLs).

9. Scalable Architecture

The application can easily support additional network services, departments, approval levels, integrations, and future enhancements without significant redesign.

10. Easy Maintenance and Customization

Since the solution is built using configurable ServiceNow components such as Service Catalog, Flow Designer, Business Rules, UI Policies, and Notifications, future modifications can be implemented quickly with minimal development effort.

Disadvantages

1. Requires ServiceNow Platform

The application depends on a ServiceNow instance and cannot operate independently without the ServiceNow platform.

2. Internet Dependency

Users require a stable internet connection to submit requests, approve requests, and monitor request progress.

3. Initial Configuration Complexity

Configuring workflows, roles, approval processes, notifications, and custom tables requires ServiceNow administration and development expertise.

4. Limited to Defined Workflow

The application processes requests according to predefined workflows. Any major changes to business processes require workflow modification and testing.

5. User Training Required

Employees, managers, and network engineers must be trained to use the Service Portal, Service Catalog, and request management features effectively.

6. Dependency on Accurate Data

Incorrect or incomplete request information may delay approvals, task assignments, or implementation activities.

7. Notification Dependency

Email notifications rely on proper ServiceNow notification configuration and email service availability.

8. Limited Offline Accessibility

The application cannot be used without network connectivity, making offline request submission and approval unavailable.

9. Customization Effort

Adding new network services, approval levels, or integrations may require additional ServiceNow development and testing.

10. Maintenance and Administration

Regular maintenance is required to monitor workflows, update catalog items, manage user roles, and ensure optimal system performance as organizational requirements evolve.

9. CONCLUSION

The **Automated Network Request Management System Using ServiceNow** successfully demonstrates how the ServiceNow platform can be utilized to automate and streamline the management of enterprise network service requests. The application replaces traditional manual processes involving emails, phone calls, and spreadsheets with a centralized, role-based solution that enables employees to submit network service requests digitally while allowing managers and network engineers to efficiently review, approve, implement, and monitor requests throughout their lifecycle.

By leveraging ServiceNow features such as **Service Catalog, scoped applications, custom tables, Business Rules, Flow Designer, Catalog UI Policies, Client Scripts, Assignment Rules, Reports, Dashboards, and Email Notifications**, the system provides a structured, transparent, and automated workflow that improves communication, reduces manual intervention, and enhances service delivery.

The project successfully achieved its primary objectives by providing a secure, scalable, and user-friendly platform for managing network service requests. Functional testing verified that

all major features—including request submission, approval workflow, automated task creation, network engineer assignment, request tracking, status updates, and automated email notifications—performed as expected. Overall, the proposed solution improves operational efficiency, enhances user experience, ensures better accountability, and provides a strong foundation for future expansion of IT service management capabilities within an organization.

10. FUTURE SCOPE

The **Automated Network Request Management System Using ServiceNow** provides a robust foundation for automating enterprise network service requests and can be further enhanced with advanced capabilities to improve operational efficiency and user experience. Future enhancements may include integration with the **Configuration Management Database (CMDB)** to automatically associate network assets and configuration items with service requests, enabling improved asset tracking and change management.

The application can also be extended with a dedicated **mobile interface**, allowing employees, managers, and network engineers to submit, approve, and monitor network requests using smartphones and tablets. Advanced **analytics dashboards** and **AI-powered reporting** can be implemented to monitor request trends, engineer workload, approval turnaround time, SLA compliance, and overall service performance.

Additional enhancements include integrating the application with **network monitoring and automation tools** to enable automatic detection of network incidents and intelligent request generation. **Predictive assignment** and **AI-assisted request categorization** can further improve task routing and resource utilization. Multi-level approval workflows can be introduced for complex or high-risk network changes, while **Microsoft Teams** or **Slack** integration can provide real-time notifications and collaboration among stakeholders.

These enhancements would make the application more intelligent, scalable, and suitable for enterprise-wide deployment while continuing to improve network service delivery, operational efficiency, governance, and the overall user experience.

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